PVsyst V8.0.19

PVsyst - Simulation report

Grid-Connected System

Project: Bergaustrasse_60

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 10.00 kWp

Berlinovo 60 - Germany

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Author

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Project summary

Geographical Site
Berlinovo 60
Germany

Situation

Latitude	52.47 °(N)
Longitude	13.48 °(E)
Altitude	33 m
Time zone	UTC+1

Project settings

Albedo	0.20

Weather data
Berlinovo 60
Meteonorm 8.2 (2001-2020), Sat=55% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 43 / 0 °

Near Shadings
no Shadings

User's needs
Daily household consumers
Seasonal modulation
Average 11.0 kWh/Day

System information

PV Array

Nb. of modules	40 units
Pnom total	10.00 kWp

Inverters

Nb. of units	1 unit
Total power	9 kWac
Pnom ratio	1.11

Results summary

Produced Energy	10736 kWh/year	Specific production	1074 kWh/kWp/year	Perf. Ratio PR	83.33 %
Used Energy	4004 kWh/year			Solar Fraction SF	56.87 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Orientation #1
Fixed plane
Tilt/Azimuth 43 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Daily household consumers
Seasonal modulation
Average 11.0 kWh/Day

PV Array Characteristics

PV module
Manufacturer Generic
Model Aleo S_81 sol / 250
(Original PVsyst database)
Unit Nom. Power 250 Wp
Number of PV modules 40 units
Nominal (STC) 10.00 kWp
Optimizer Array 2 string x 10 In series
At operating cond. (50°C)
Pmpp 9.01 kWp
U mpp 480 V
I mpp 19 A

Inverter
Manufacturer Generic
Model 9 kWac inverter
(Original PVsyst database)
Unit Nom. Power 9.00 kWac
Number of inverters 2 * MPPT 50% 1 unit
Total power 9.0 kWac
Operating voltage 150-750 V
Max. power (=>25°C) 10.00 kWac
Pnom ratio (DC:AC) 1.11
No power sharing between MPPTs

AdvanSol Optimizer
Model APT-MC-MRO-updated
Unit Nom. Power 800 W
Input modules 2 in series

Total PV power
Nominal (STC) 10 kWp
Total 40 modules
Module area 57.4 m²
Cell area 49.8 m²

Total inverter power
Total power 9 kWac
Number of inverters 1 unit
Pnom ratio 1.11

Array losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 430 mΩ
Loss Fraction 1.50 % at STC

LID - Light Induced Degradation
Loss Fraction 2.00 %

Module Quality Loss
Loss Fraction -0.50 %

Module mismatch losses
Loss Fraction 0.50 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

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Array losses

Spectral correction

FirstSolar model

Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0,85914	-0,02088	-0,0058853	0,12029	0,026814	-0,001781

System losses

Unavailability of the system

Time fraction 2.0 %
7.3 days,
3 periods

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Detailed User's needs

Daily household consumers, Seasonal modulation, average = 11.0 kWh/day

Summer (Jun-Aug)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	10	10/lamp	5.0	500
TV / PC / Mobile	2	100/app	5.0	1000
Domestic appliances	1	500/app	4.0	2000
Fridge / Deep-freeze	2		24	1598
Dish- & Cloth-washers	1		2	2000
Ventilation	1	100 tot	24.0	2400
Air conditioning	1	1000 tot	3.0	3000
Stand-by consumers			24.0	144
Total daily energy				12642

Autumn (Sep-Nov)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	10	10/lamp	5.0	500
TV / PC / Mobile	2	100/app	5.0	1000
Domestic appliances	1	500/app	5.0	2500
Fridge / Deep-freeze	2		24	1598
Dish- & Cloth-washers	1		2	2000
Ventilation	1	100 tot	24.0	2400
Stand-by consumers			24.0	144
Total daily energy				10142

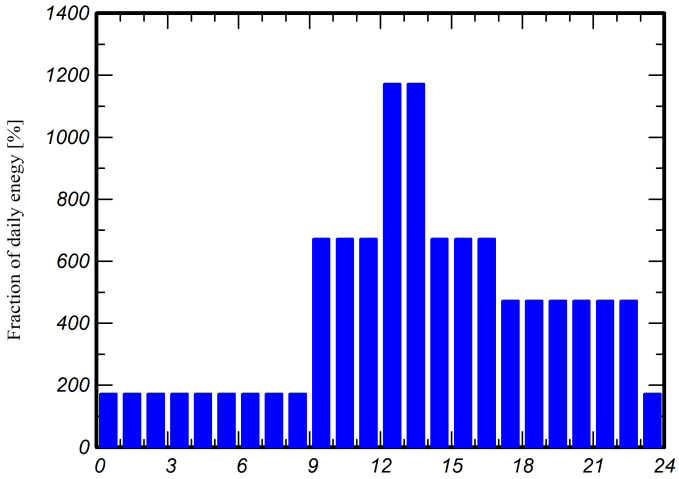
Winter (Dec-Feb)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	10	10/lamp	6.0	600
TV / PC / Mobile	2	100/app	6.0	1200
Domestic appliances	1	500/app	6.0	3000
Fridge / Deep-freeze	2		24	1598
Dish- & Cloth-washers	1		2	2000
Ventilation	1	100 tot	24.0	2400
Stand-by consumers			24.0	144
Total daily energy				10942

Spring (Mar-May)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	10	10/lamp	5.0	500
TV / PC / Mobile	2	100/app	5.0	1000
Domestic appliances	1	500/app	5.0	2500
Fridge / Deep-freeze	2		24	1598
Dish- & Cloth-washers	1		2	2000
Ventilation	1	100 tot	24.0	2400
Stand-by consumers			24.0	144
Total daily energy				10142

Hourly distribution



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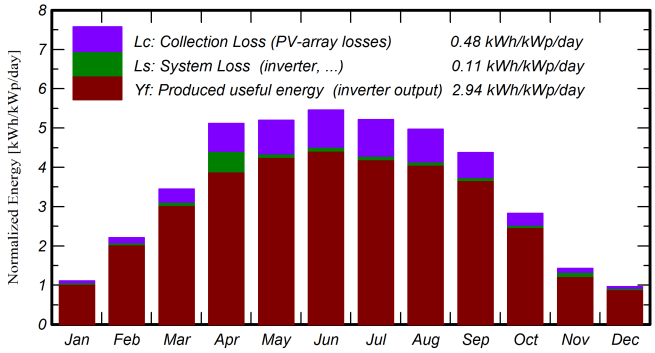
Main results

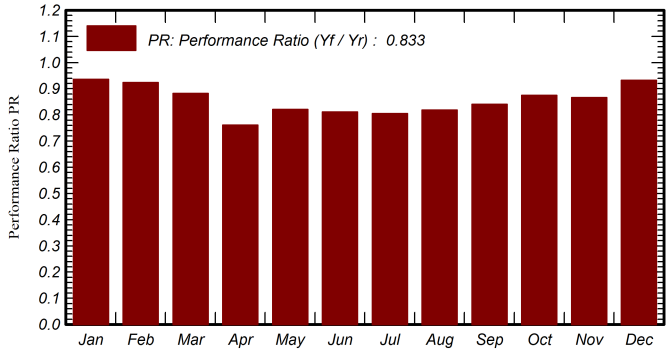
System Production

Produced Energy	10736 kWh/year	Specific production	1074 kWh/kWp/year
Used Energy	4004 kWh/year	Perf. Ratio PR	83.33 %
		Solar Fraction SF	56.87 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	19.6	14.77	-0.28	34.3	32.6	331	339	129	192	210
February	37.3	23.32	0.82	61.9	58.8	584	306	153	418	153
March	76.4	40.02	3.86	106.8	101.1	967	314	180	762	134
April	127.4	53.98	9.00	153.4	144.7	1323	304	179	990	125
May	160.5	73.09	14.59	161.0	151.2	1350	314	219	1103	96
June	170.5	78.32	17.17	163.6	153.7	1357	379	287	1042	93
July	165.9	77.51	19.72	161.7	151.8	1331	392	288	1015	104
August	142.4	74.65	19.11	154.0	144.7	1286	392	283	978	109
September	97.5	41.98	14.22	131.2	123.8	1125	304	189	913	115
October	54.9	28.83	9.32	87.7	83.2	784	314	169	599	146
November	23.4	15.18	4.43	42.7	40.6	400	304	106	264	198
December	14.5	9.98	0.46	29.9	28.4	287	339	95	183	244
Year	1090.2	531.64	9.42	1288.4	1214.7	11126	4004	2277	8459	1727

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	E_Grid	Energy injected into grid
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

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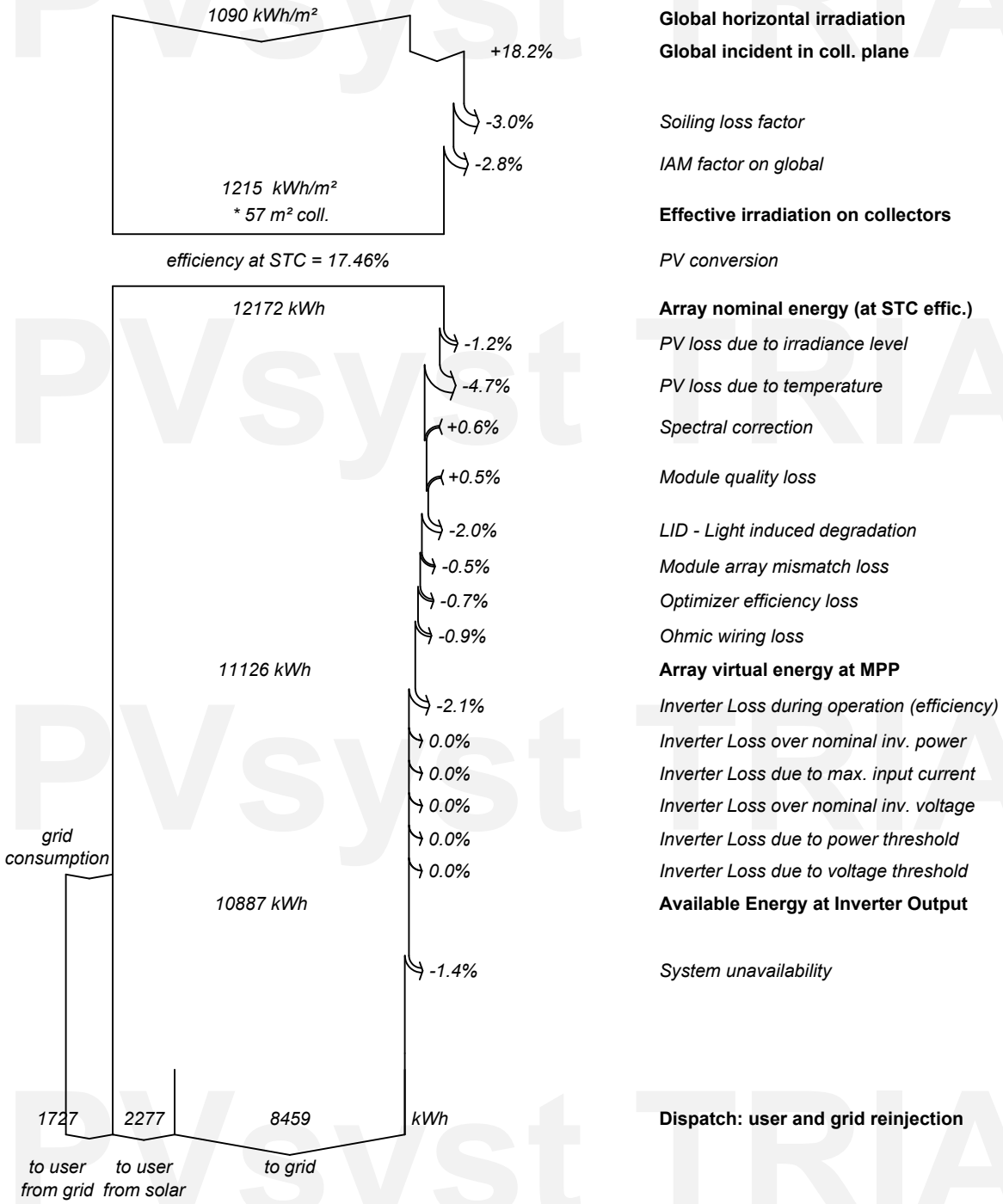
Loss diagram



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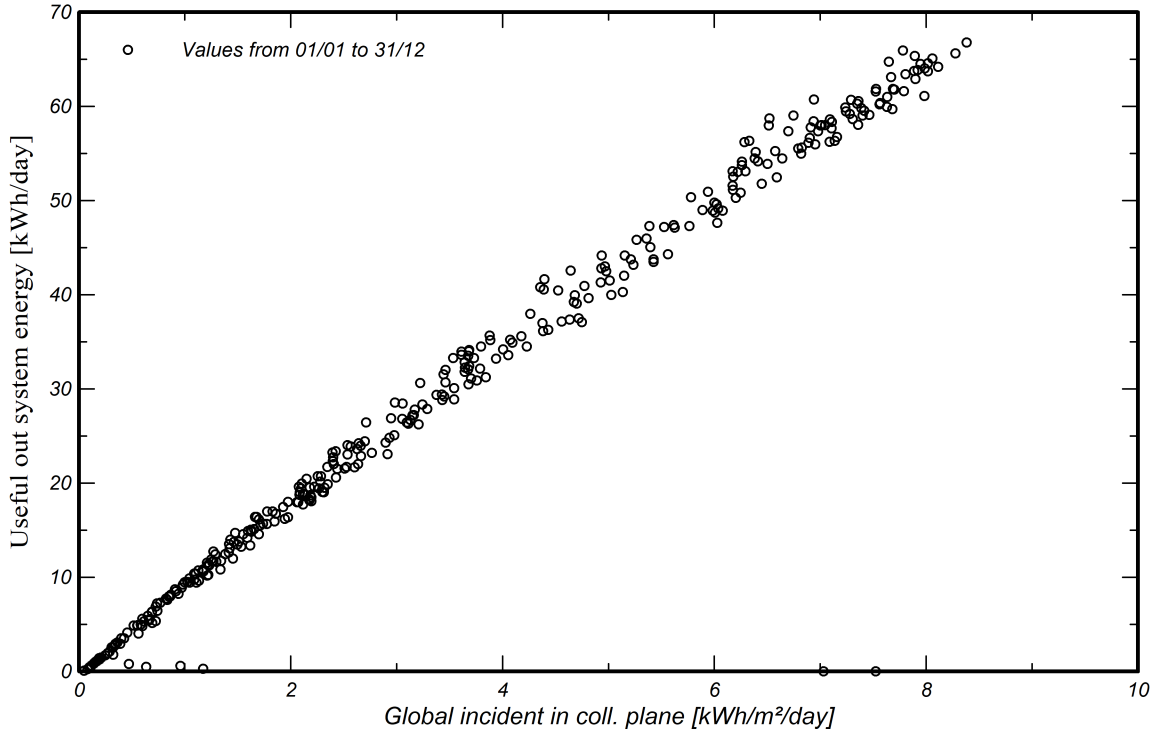
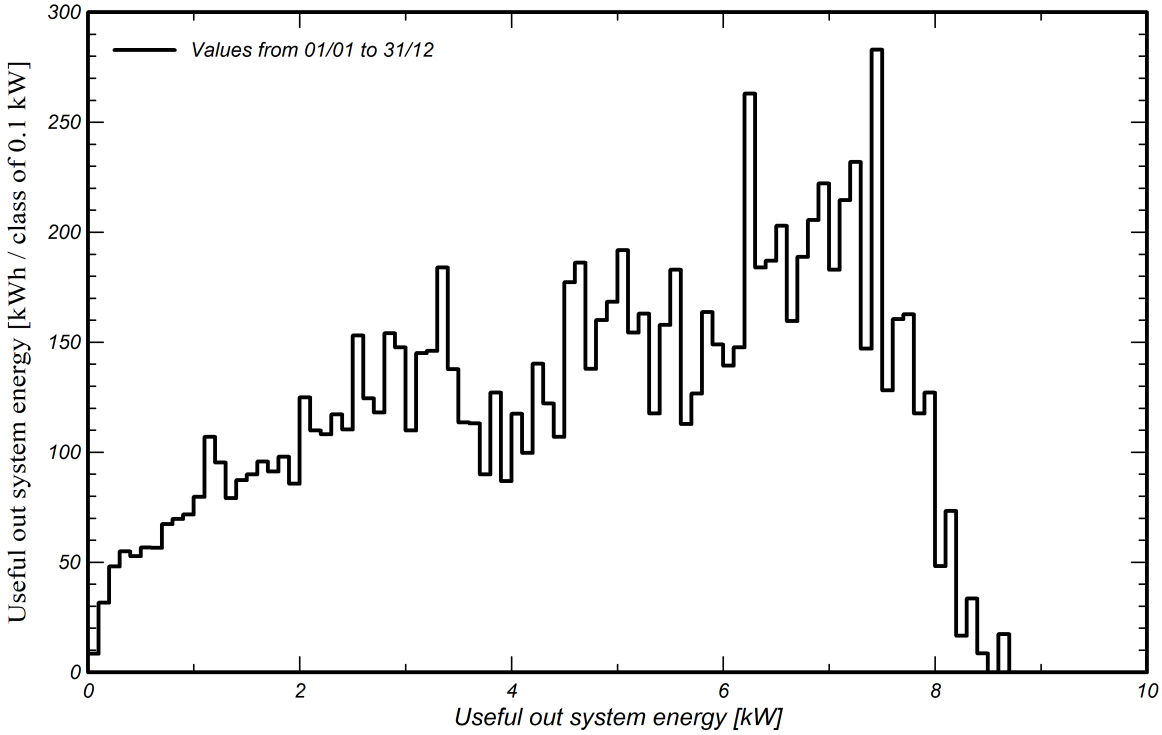
Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



Single-line diagram

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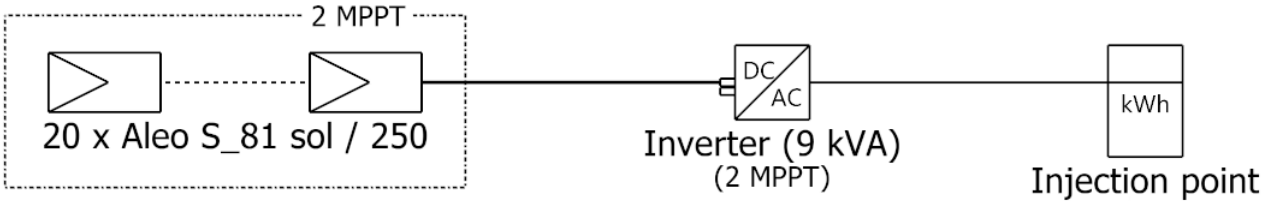

PV module	Aleo S_81 sol / 250
Inverter	9 kWac inverter
String	20 x Aleo S_81 sol / 250

Bergaustrasse_60

VC0 : New simulation variant

27/01/26

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Cost of the system

Installation costs

Item	Quantity units	Cost EUR	Total EUR
		Total	0.00
		Depreciable asset	0.00

Operating costs

Item	Total EUR/year
Total (OPEX)	0.00

System summary

Total installation cost	0.00 EUR
Operating costs	0.00 EUR/year
Useful energy from solar	2277 kWh/year
Energy sold to the grid	8459 kWh/year
Cost of produced energy (LCOE)	0.0000 EUR/kWh

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CO_2 Emission Balance

Total: 112.1 tCO_2

Generated emissions

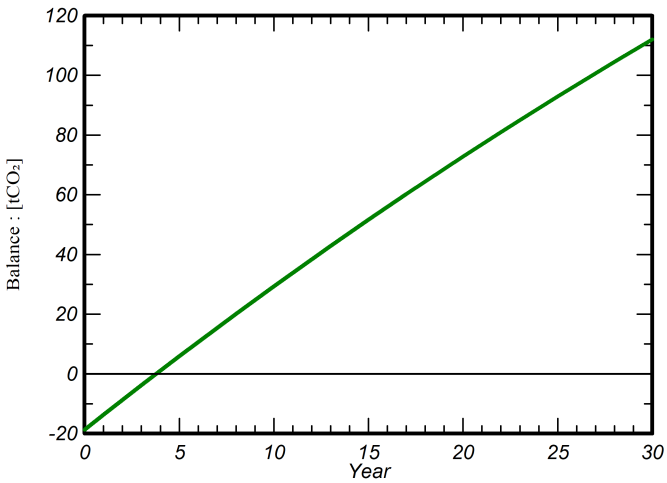
Total: 18.69 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total:	150.7 tCO_2
System production:	10.74 MWh/yr
Grid Lifecycle Emissions:	468 gCO_2/kWh
Source:	IEA List
Country:	Germany
Lifetime:	30 years
Annual degradation:	1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	10.00 kWp	17127
Supports	3.12 kgCO2/kg	400 kg	1249
Inverters	309 kgCO2/	1.00	309

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